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Research Paper

Performance analysis of a He-Xe Brayton cycle based on recuperator configuration for nuclear space power systems

Junbin Zhou [a], Liyan Cao [b], Gaoliang Liao [a], Feng Zhang [a,*]

[a] College of Mechanical and Vehicle Engineering, Hunan University, Changsha 410082, China
[b] Science and Technology on Reactor System Design Technology Laboratory, Nuclear Power Institute of China, Chengdu 610213, China

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ABSTRACT

This study investigates the optimization of a helium-xenon (He-Xe) Brayton cycle recuperator for space power systems. Through parametric analysis and multi-objective optimization (NSGA-II), the impact of internal geometric parameters—channel diameters and counts on the hot and cold sides—on system mass, efficiency, and pressure loss is evaluated. The results show that the intercooled cycle outperforms the simple regenerative cycle in efficiency, while its total mass is slightly higher. A key finding is the trade-off in recuperator design: minimizing channel dimensions reduces mass but increases pressure loss, particularly on the hot side where low-density gas flow dominates. Consequently, a configuration with smaller and fewer channels on the cold side is proposed to balance this trade-off. The optimal solutions are selected from the Pareto front using the TOPSIS method, providing clear design guidelines for lighter and more efficient space power systems.

1. Introduction

With the rapid development of space technology, its demand for energy is also increasing day by day. Different from ground power systems, considering the high cost of space transportation, space energy systems, while ensuring high cycle efficiency, must also strive to minimize the total system mass [1]. Nuclear reactors, due to their compact size, light weight, long lifespan, and high energy density characteristics, have become an ideal energy source for space exploration [2,3]. The closed Brayton cycle, relying on its high thermal efficiency and low mass per unit power, is an ideal energy conversion technology matched with nuclear reactors [4]. Therefore, gas-cooled space reactors employing direct Brayton cycles have become one of the highly promising directions for future space energy systems [5]. The gas-cooled reactors typically choose helium-xenon (He-Xe) mixed gas or supercritical carbon dioxide (S-CO_2) as the cycle working fluid.

Currently, a large number of studies have focused on the application of S-CO_2 in gas-cooled reactor. Xue et al. [6] conducted thermodynamic performance and total system mass analysis on an S-CO_2 Brayton cycle direct-cooled reactor system, pointing out that at a core outlet temperature of 600 °C, 16 MPa is the optimal operating pressure. Zhang et al. [7] introduces a 15 MW gas-cooled reactor supercritical carbon dioxide Brayton cycle and optimized the core configuration through multi-physics modeling. Gao et al. [8] adopts an optimization strategy combining machine learning and evolutionary algorithms to efficiently identify the optimal configuration among several typical supercritical carbon dioxide cycles. Also developed thermodynamic and mass analysis to investigate the influence of design variables on system performance. In addition to thermodynamics and mass, the influence of design variables on system performance can also be examined from the perspectives of environmental and economic performance [9]. Yuan et al. [10] designed a 6 MW supercritical carbon dioxide Brayton cycle power generation system and established thermodynamic calculation models and mass evaluation models. Used a genetic algorithm to perform global multi-objective optimization; the results indicate that the recompression cycle achieves the highest cycle efficiency, while the simple regenerative cycle exhibits the lowest cycle efficiency. *Ma* et al. [11] enhanced Brayton cycle performance by altering the critical point through mixing other gases with CO_2 while employing a nondominated sorting genetic algorithm to simultaneously optimize cycle efficiency and total cycle cost.

However, in space application scenarios, the higher operating pressure of S-CO_2 may cause corrosion of components such as piping, therefore the tendency is usually to use inert gas working fluids. Existing research has fully demonstrated the feasibility of helium-xenon mixed gas as a cycle working fluid for space nuclear power systems. The genetic algorithm for global multi-objective optimization is a traditional

* Corresponding author,
E-mail address: fengzhang@hnu.edu.cn (F. Zhang).

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Nomenclature

A	area, m^2
c_p	specific heat capacity at constant pressure, $J{\cdot}kg^{-1}{\cdot}K$
D	diameter, m
f	friction factor
h	specific enthalpy, $kJ{\cdot}kg^{-1}$
H	height, m
HTC	heat transfer coefficient, $W{\cdot}m^{-2}{\cdot}K^{-1}$
L	length, m
m	mass flow rate, $kg{\cdot}s^{-1}$
M	Weight, kg
NTU	Number of Transfer Units
Nu	Nusselt number
P	pressure, kPa
Pr	Prandtl number
q	line power density, $kW{\bullet}m^{-1}$
Q	heat transfer rate, kW
Re	Reynolds number
t	thickness, mm
T	temperature, K
u	flow velocity, $m{\bullet}s^{-1}$
U	total heat transfer coefficient, $W{\cdot}m^{-2}{\cdot}K^{-1}$
V	volume, m^3
W	power, kW

Greek letters

λ	thermal conductivity, $W{\cdot}m^{-1}{\cdot}K^{-1}$
E	efficient of the sub-exchanger, %
η	efficiency, %
ρ	density, kg/m^3
δ	average heat conduction distance, mm

Subscripts

1, 2, et al.	state points
cold	cold fluid
hot	cold fluid
in	inlet
out	outlet
max	maximum
min	minimum
net	net power

optimization method. Mohamed S et al. employed weighting coefficients to balance thermodynamic performance and total product unit cost, enabling customized system performance for different objectives and ensuring a comprehensive optimization process [12]. Compared to pure helium (He), adding a small amount of xenon (Xe) can increase the average molar mass of the mixed gas, thereby effectively reducing the load on turbo machinery [13]. Cheng et al. [14] compared Brayton cycles using different working fluids. The results demonstrated that under ultra-high operating temperatures, the He-Xe Brayton cycle exhibits a more significant increase in power compared to the SCO_2 Brayton cycle. Guan et al. [15] conducted an investigation on the effects of average molar mass of He-Xe mixtures on the efficiency and mass of He-Xe Brayton cycle for gas-cooled reactor. The results showed that although a lower molar mass of He-Xe mixtures was beneficial to reduce system volume, it reduces system efficiency. Tong et al. [16] proposed a multi-stage compression He-Xe Brayton cycle and established thermodynamic and mass models. Separately discussed the effects of split ratio, turbine inlet temperature, heat exchanger effectiveness, etc. on the cycle system's thermal efficiency, total mass, and specific power. Also performed multi-objective optimization with the system's thermal efficiency, total mass, and specific power as optimization objectives. Tang et al. [17] developed a thermodynamic model for a simple regenerative He-Xe Brayton cycle to investigate the influence of key parameters on system performance. The results show that an increase in the helium mass fraction in the He-Xe working fluid leads to changes in physical properties, consequently causing an increase in turbine output work, cycle power output, and cycle efficiency. Zhang et al. [18] developed a Modelica-based one-dimensional thermodynamic cycle library for system-level modeling and multi-objective optimization of a helium-xenon closed Brayton cycle. Parth Prajapati et al. conducted a multi-objective optimization study on a combined Brayton and inverse Brayton cycle. This assists users in selecting operating parameters according to requirements and provides multiple solutions [19].

Apart from the thermodynamic performance of the He-Xe Brayton cycle, the total system mass is another key design parameter for space energy systems. However, due to the close interrelation between internal system components and conflicts between optimization objectives (such as pursuing high efficiency usually requiring lower mass), simultaneously achieving high efficiency and low mass is often challenging. For example, increasing the ratio of the highest to the lowest cycle temperature helps improve thermoelectric conversion efficiency [20], but increasing the highest cycle temperature, under given fuel peak temperature limitations, will lead to a reduction in reactor power density [21]. To this end, Jiang et al. [22] proposed a new mass estimation model and conducted dual-objective optimization research on efficiency and mass to enhance overall system performance. In helium-xenon Brayton cycles, most research on thermal efficiency and mass optimization targets space nuclear reactors. In such systems, heat exchangers or radiators occupy a considerable proportion of the mass [23]. For space nuclear reactors, due to neutronic constraints, the size of the reactor will not be affected by the power level [24].

Printed Circuit Heat Exchangers (PCHEs) have significant advantages as recuperators and coolers in Brayton cycles; their compact micro channel structure can achieve extremely high heat exchange efficiency and power density, greatly reducing equipment volume and weight [25–27]. In Brayton cycles, the recuperator mass serves as the dominant contributor to variations in the system total mass [28]. Shi et al. [29] investigated the effects of three novel internal vortex generators (rectangular annular ribs, Kagome lattice, and body-centered cubic (BCC) lattice) on heat transfer enhancement in helium-xenon mixture via numerical analysis. The results showed that the induced vortex system significantly improved heat transfer capability, increasing the overall Nusselt number to approximately 2. 6 times that of a smooth tube while maintaining an acceptable pressure drop. Wang et al. [30] studied the flow and heat transfer characteristics of He-Xe mixture under natural circulation conditions. A three-dimensional model of a natural circulation loop was established using numerical calculation methods. Yin et al. [31] examined the thermal-hydraulic performance of helium-xenon within a PCHE. As reviewed above, although many researches have been paid attention to the effect of the thermodynamic parameters on the thermodynamic performance and mass of the He-Xe Brayton cycle for gas-cooled reactor, rare investigations focused on the effect of the design of the heat exchangers on the system performance and mass.

This study systematically investigates the influence of the recuperator's internal structure on the performance of the helium-xenon Brayton cycle. The work is organized as follows: Section 2 introduces two distinct helium-xenon Brayton cycles; Section 3 presents the thermodynamic models of each component along with their validation. In Section 4, the variation ranges of four geometric parameters-channel diameter (2–4 mm) and channel number (6000–8000) are considered, focusing on key thermodynamic parameters and system mass characteristics, while accounting for actual component pressure losses under

different operating conditions. Based on the analysis results, a multi-objective optimization is conducted for these four variable parameters with the aim of balancing cycle efficiency and total system mass, leading to the identification of an optimal set of design parameters. Finally, a summary of the work is provided in Section 5.

2. System layout description

In this study, the traditional recuperated He-Xe Brayton cycle and the intercooling He-Xe Brayton cycle coupled with the gas-cooled reactor are employed for the space power generation, and the layout of two systems are shown in Fig. 1. The recuperated He-Xe Brayton cycle

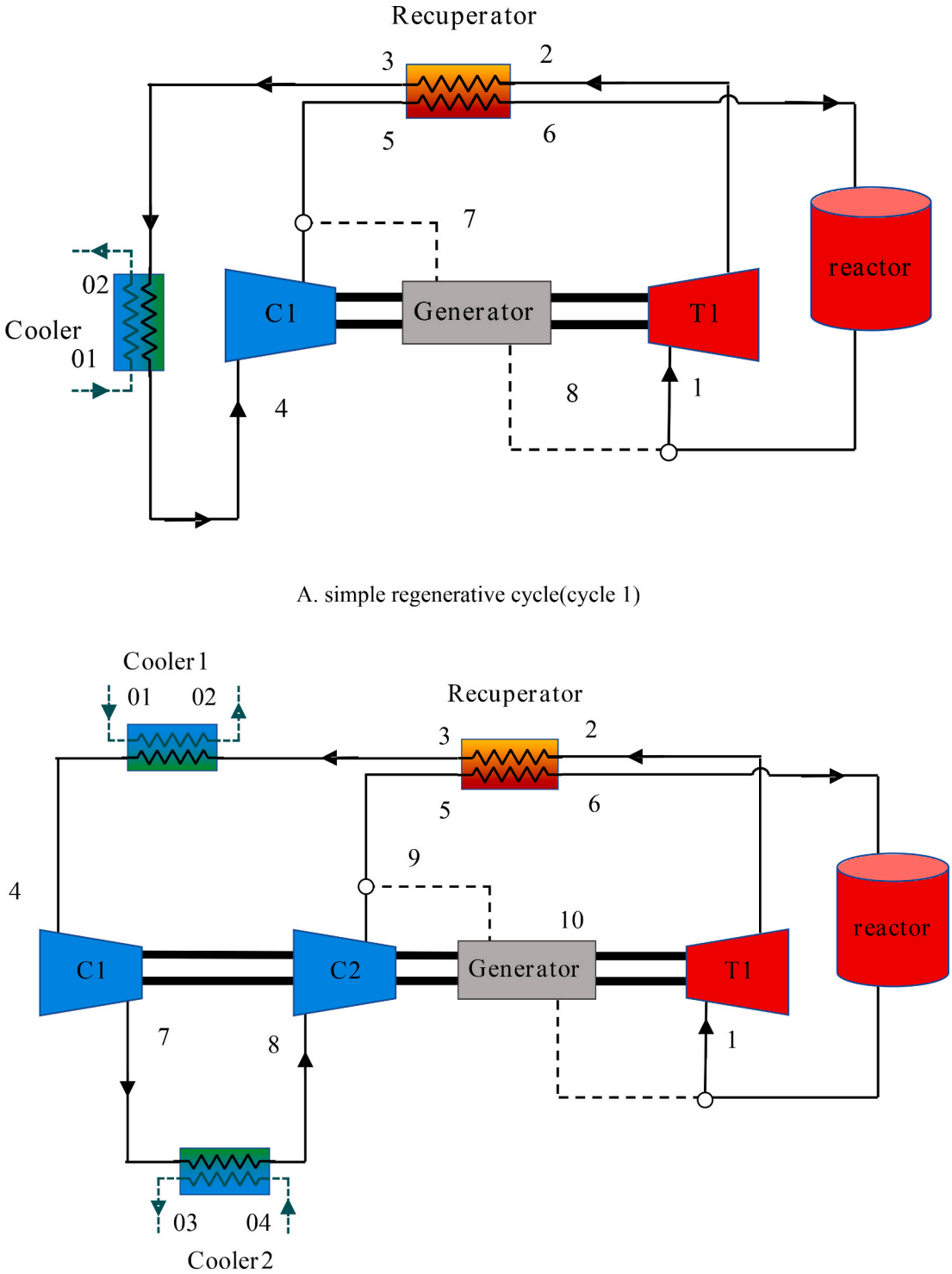


Fig. 1. He-Xe Brayton cycle.

coupled with the gas-cooled reactor consists of a turbine, a generator, a compressor, a recuperator, a cooler and the gas-cooled reactor. The He-Xe mixture with low temperature is compressed by the compressor driven by the turbine(4–5). After compression, the high-pressure He-Xe mixture is split into two streams. One stream passes sequentially through the recuperator in which it recovers heat from the turbine exhaust(5–6) and gas-cooled reactor in which it absorbs heat from fuel rods to avoid exceeding the material limited temperature(6–1). Afterwards, the high-temperature He-Xe mixture expands in the turbine to convert the thermal energy into mechanical energy, and the turbine drives the electric generator to produce the electricity(1–2). Meanwhile, the other stream flows through the generator to cool the generator, and then it recombines with the turbine exhaust(7–8-1). The combined flow then flows through the recuperator to release heat(2–3). Afterwards, it passes the cooler to transfer heat to the liquid NaK alloy, and meanwhile the liquid NaK alloy releases heat to space via radiators(3–4). Finally, the low-temperature and low-pressure He-Xe mixture returns. It should be also noted that both the compressor and generator are directly driven by the turbine, which is beneficial to a more compact layout. The intercooling He-Xe Brayton cycle operates similarly to the recuperated He-Xe Brayton cycle, but the former incorporates intercooling process between two compressors to reduce compressor work. Fig. 2 illustrates the computational flow for Cycle 1. It should be noted that for Cycle 2, the pressure ratios of the two compressors are identical.

3. Mathematical models

The system model was developed in MATLAB. The power cycle comprises compressors, turbines, recuperators, coolers, and heaters (gas-cooled reactor). Thermodynamic properties of the He-Xe mixture

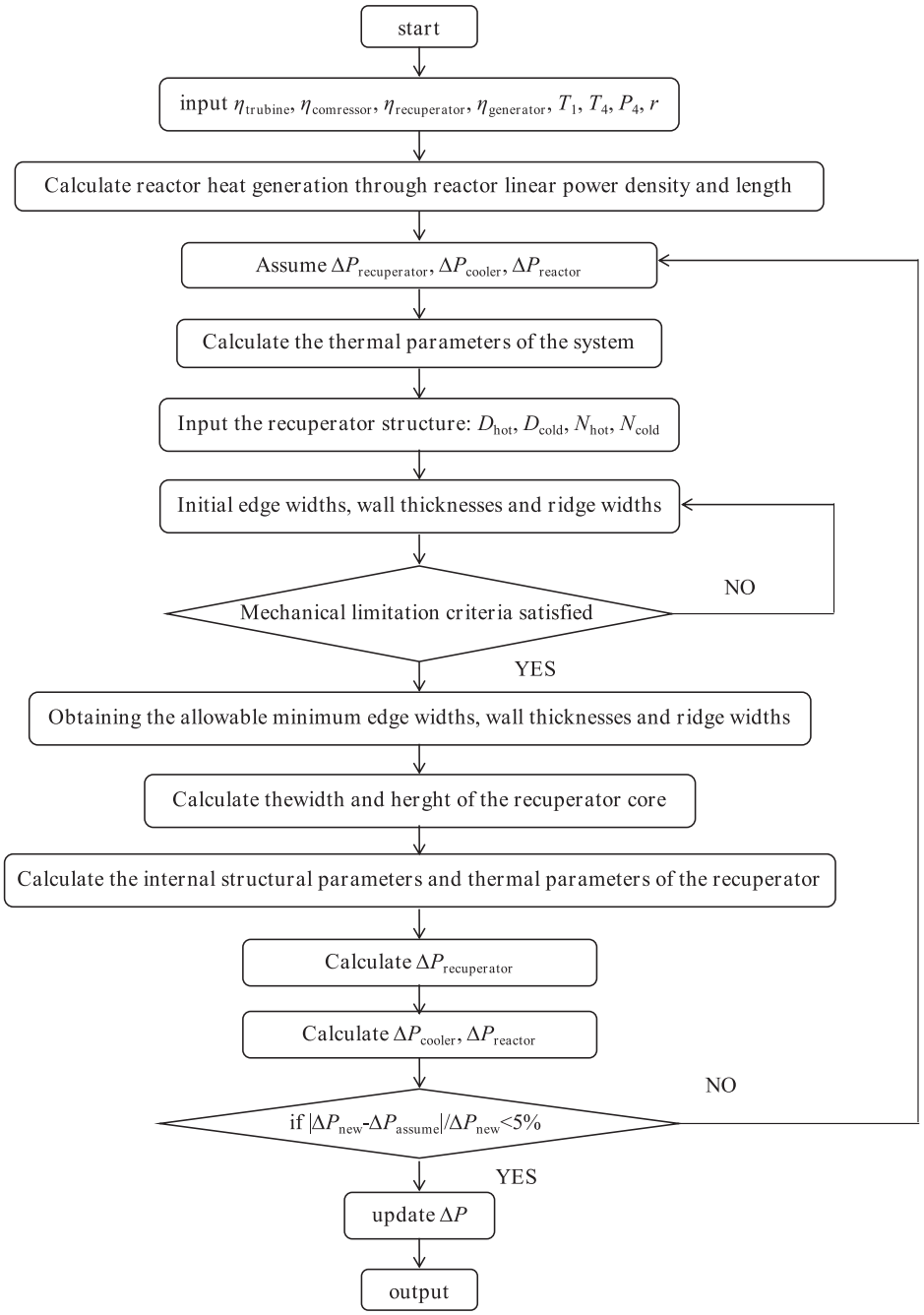


Fig. 2. Flow chart of He-Xe Brayton cycle.

follow Tournier and El-Genk's established model. [37]

In this study, some assumptions are employed to simplify the mathematical modeling, as given below.

(1) The He-Xe power cycle is assumed to operate under steady state.
(2) The heat losses in the pipelines and heat exchangers are neglected.
(3) There is no leakage of gas in the He-Xe power cycle.

3.1. Gas-cooled reactor model

The gas-cooled reactor employs a hexagonal design utilizing modular graphite blocks. As shown in Fig. 3, these blocks contain dedicated coolant channels and fuel channels arranged in a triangular lattice configuration, with coolant channels positioned interstitially between fuel elements. The cooling system utilizes direct-contact heat absorption with He-Xe.

The reactor core consists of multiple fuel rods. Under the simplified assumption of uniform linear power density across all fuel rods, and considering the large number of flow channels, computational efficiency is prioritized by calculating,

$$Q_{\text{reactor}} = c_{\text{p}}\dot{m}_{\text{He-Xe}}(T_{\text{out}} - T_{\text{in}}) \tag{1}$$

where $Q_{\text{reactor}}$is total thermal power output of the reactor, $c_{\text{p}}$is specific heat capacity of He-Xe mixture, $m_{\text{He-Xe}}$is total mass flow rate of He-Xe, $T_{\text{in}}$is coolant inlet temperature of gas-cooled reactor, T_{out} is coolant outlet temperature of gas-cooled reactor.

The gas-cooled reactor is subdivided into multiple axial segments. The outlet temperature is determined by calculating the temperature rise through each segment sequentially, as detailed in Fig. 4. The coolant temperature distribution within the reactor core is governed by,

$$T_{\text{out,i}} = T_{\text{in}} + \frac{Q_{\text{reactor,i}}}{c_{\text{p,i}}m_{\text{He-Xe}}} \tag{2}$$

The volumetric heat generation rate ($Q_{\text{reactor,i}}$) follows a cosine distribution along the axial direction, as given below,

$$Q_{\text{reactor,i}} = \int_{-\frac{H}{2}}^{zi} q_{l,\max}\cos\left(\frac{\pi z}{L}\right)dz \tag{3}$$

where the $q_{l,\max}$is peak linear power density of fuel rods, H is height of the gas-cooled reactor core, z is axial position coordinate for segment i.

The frictional pressure drop along the hot channel is calculated using the following formula [32],

$$\Delta P_{f,m} = f_i\frac{L_i}{d_i}\frac{\rho_i u_i^2}{2} \tag{4}$$

where the f is Darcy friction factor, ρ and u is fluid density and flow velocity, L is channel length of the sub-exchanger, d is hydraulic diameter of the channel. Nu and f can be calculated by [32],

$$\begin{cases} Nu = 4.089 \quad \text{Re} < 2300 \\ Nu = 4.089 + (Nu_{5000} - 4.089)\cdot(\text{Re} - 2300)/(5000 - 2300) \qquad 2300 \le \text{Re} < 5000 \\ Nu = (f/8)(\text{Re} - 1000)\text{Pr}\Big/\left(1 + 12.7(f/8)^{1/2}\left(\text{Pr}^{2/3} - 1\right)\right) \qquad \text{Re} \ge 5000 \end{cases} \tag{5}$$

$$f = \begin{cases} \dfrac{64}{\text{Re}}, \text{Re} \le \text{Re}_0 \\ 4.4\text{Re}^{-0.0595}e^{\frac{0.00275}{\Delta}}, \text{Re}_0 \le \text{Re} \le \text{Re}_1 \\ (f_2 - f_1)e^{-[0.0017(\text{Re}_2 - \text{Re})]^2 + f_1}, \text{Re}_1 \le \text{Re} \le \text{Re}_2 \\ \dfrac{1}{\left[2\log\left(\frac{2.51}{\text{Re}\sqrt{f}} + \frac{\Delta}{3.7}\right)\right]^2}, \text{Re} \ge \text{Re}_2 \end{cases} \tag{6}$$

where Δ is the channel relative roughness, f_1, and f_2 are auxiliary parameters, and Re_0, Re_1, and Re_2 are the Reynolds numbers defining the flow points at the end of the laminar flow, the beginning of the transition regime,and the beginning of the turbulent flow, respectively. These are expressed as:

$$f_1 = \begin{cases} 0.032, \Delta \le 0.007 \\ 0.0733 - \dfrac{0.0109}{\Delta^{0.286}}, \Delta > 0.007 \end{cases} \tag{7}$$

$$f_2 = \frac{1}{\left[2\log\left(\frac{2.51}{\text{Re}\sqrt{f_2}} + \frac{\Delta}{3.7}\right)\right]^2} \tag{8}$$

$$\text{Re}_0 = \begin{cases} 2000, \Delta \le 0.007 \\ 754e^{\frac{0.0065}{\Delta}}, \Delta > 0.007 \end{cases} \tag{9}$$

$$\text{Re}_1 = \begin{cases} 2000, \Delta \le 0.007 \\ \dfrac{1165}{\Delta}, \Delta > 0.007 \end{cases} \tag{10}$$

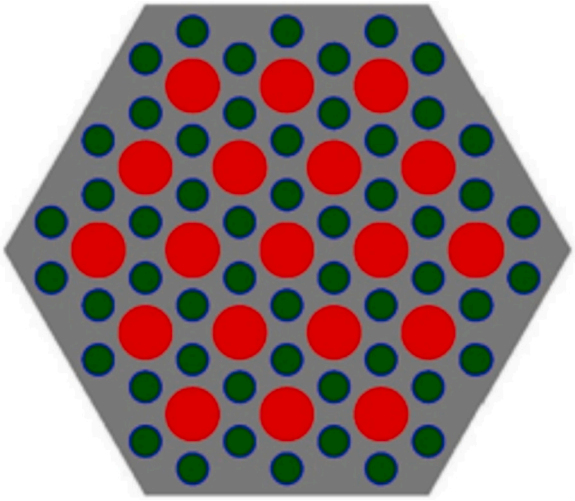


Fig. 3. The structure of gas-cooled reactors.

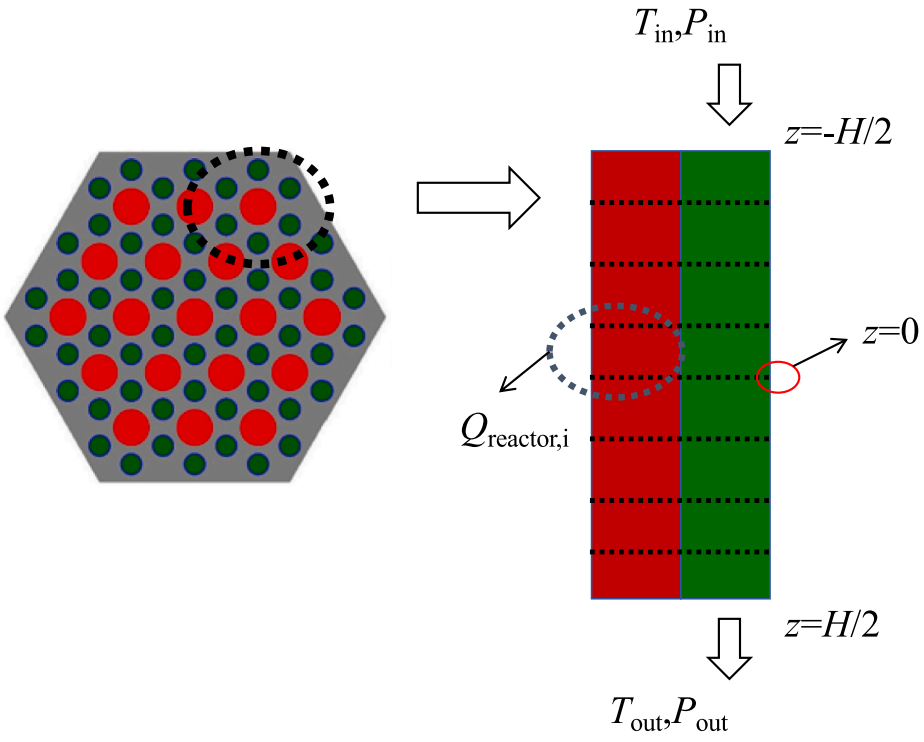


Fig. 4. The thermal model of gas-cooled reactors.

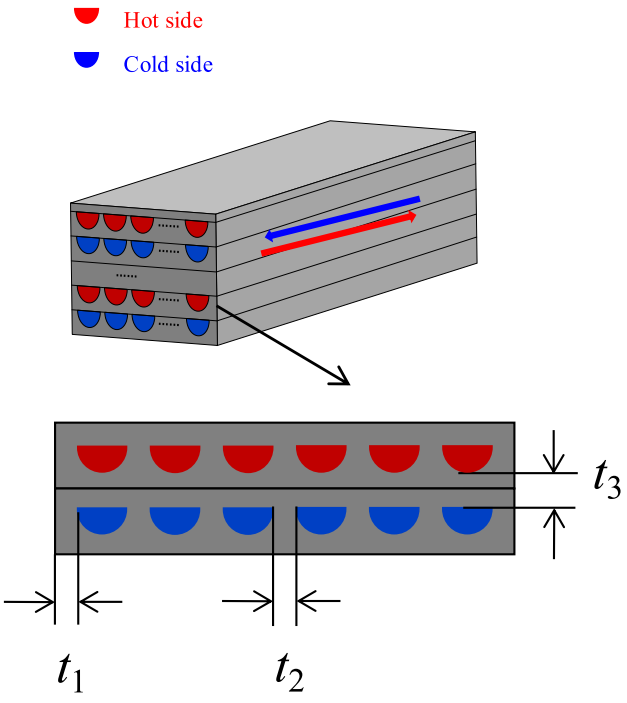


Fig. 5. The structure of recuperator.

$$\mathrm{Re}_2 = 2090\left(\frac{1}{\Delta}\right)^{0.0635} \tag{11}$$

3.2. Heat exchanger model

Within the power conversion system, the heat exchanger assembly comprises a recuperator and a cooler. The recuperator transfers heat between counter flowing streams of He-Xe gas on both sides, while the cooler facilitates thermal exchange between NaK-78 (22% Na, 78% K) and the He-Xe working fluid. The most frequently used thermo physical properties of NaK-78 have been summarized as polynomial formulations in the data sheet [33].

The heat exchangers employ semi-circular straight-channel PCHEs, as illustrated in Fig. 5, where t_1, t_2, t_3 denote the ridge width, wall thickness, and edge width, respectively. The parameters t_1, t_2, t_3 are determined through a structural strength model developed in our previous publication [34]. Based on the calculated values of t_1, t_2, t_3, the width and height of the recuperator can be derived, thereby yielding the total cross-sectional area of the recuperator.

Heat exchanger effectiveness (ε), defined as the ratio of actual heat transfer rate to the maximum theoretically possible heat transfer rate, can be expressed as below [35],

$$\varepsilon = \frac{\dot{m}_c \bar{c}_{p,c}\left(T_{\mathrm{cold,out}} - T_{\mathrm{cold,in}}\right)}{C_{\min}\left(T_{\mathrm{hot,in}} - T_{\mathrm{c,out}}\right)} \tag{12}$$

where the $C_{\min}$ is minimum heat capacity rate of recuperator, the $\bar{c}_{p,c}$ is the Average Specific Heat at Constant Pressure.

Heat transfer area serves as a critical metric for evaluating the size and economic viability of heat exchangers, while also constituting a fundamental structural parameter. As shown in Fig. 6. This study employs a discretized sub-exchanger methodology to compute the heat transfer area.

$$Q_i = U_i A_i \Delta T_m^i \tag{13}$$

where λ_i is Log Mean Temperature Difference (*LMTD*) of the sub-exchanger, $U_i$is Overall heat transfer coefficient (U) of the sub-exchanger, $A_i$is the heat transfer area of the sub-heater sub-exchanger.

$$\frac{1}{U_i A_i} = \frac{1}{h_h^i A_h^i} + \frac{t}{\lambda_s A_c^i} + \frac{1}{h_c^i A_c^i} \tag{14}$$

$$\Delta T_m^i = \frac{\Delta T_{\max}^i - \Delta T_{\min}^i}{\ln \frac{\Delta T_{\max}^i}{\Delta T_{\min}^i}} \tag{15}$$

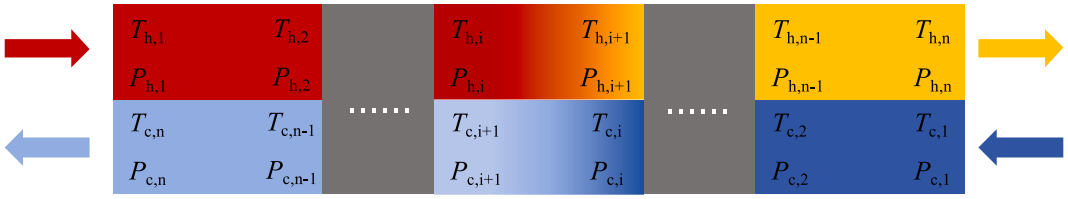


Fig. 6. Schematic diagram of discrete model for PCHE.

where the t is the thickness of the metal material, the$\lambda_s$is the thermal conductivity of the metal material, the heat transfer coefficient (*HTC*) can be calculated by,

$$h_i = Nu_i\lambda_i/d_i \tag{16}$$

The calculation methods for *Nu* and *f* of Xe-He gas follow conventional reactor coolant correlations, whereas the *Nu* for liquid metal employs the Seban-Shimazaki correlation [36].

$$Nu = 5.0 + 0.025(\mathrm{RePr})^{0.8} \tag{17}$$

Number of Transfer Units (*NTU*) can be calculated by,

$$NTU = \begin{cases} \dfrac{\ln\left(\dfrac{1-\varepsilon C_R}{1-\varepsilon}\right)}{1-C_R} & C_R \neq 1 \\ \dfrac{\varepsilon}{1-\varepsilon} & C_R = 1 \end{cases} \tag{18}$$

where the $C_R$is the ratio of the minimum heat capacity to the maximum heat capacity , the$\varepsilon$is the effectiveness of the sub-exchanger.

$$C_R = C_{\min}/C_{\max} \tag{19}$$

$$\varepsilon = \frac{Q}{C_{\min}\left(T_{\mathrm{hot,in}} - T_{\mathrm{cold,in}}\right)} \tag{20}$$

The *UA* of the heat exchanger is the product of the number of heat transfer units and the minimum heat capacity.

$$UA_i = NTU \cdot C_{\min} \tag{21}$$

$$UA = \sum_{1}^{n} UA_i \tag{22}$$

3.3. Compressor and turbine model

The power output of the turbine or the power consumption of the compressor can be calculated using the inlet/outlet temperatures and specific heat capacities through the following thermodynamic relationships:

$$\begin{cases} W_{\mathrm{compressor}} = \bar{c}_p \dot{m}(T_{\mathrm{out}} - T_{\mathrm{in}}) \\ W_{\mathrm{turbine}} = \bar{c}_p \dot{m}(T_{\mathrm{in}} - T_{\mathrm{out}}) \end{cases} \tag{23}$$

3.4. Evaluative criteria

Net power output (W_{net}), thermal efficiency (η), and total system mass (M_{cycle}) serve as primary evaluation criteria for the He-Xe Brayton cycle. These parameters are defined as follows:

$$W_{\mathrm{net}} = W_{\mathrm{trubine}} - W_{\mathrm{compressor}} \tag{24}$$

$$\eta = \frac{W_{\mathrm{net}}}{Q_{\mathrm{reactor}}} \tag{25}$$

$$M_{\mathrm{sys}} = M_{\mathrm{trubine}} + M_{\mathrm{compressor}} + M_{\mathrm{cooler}} + M_{\mathrm{recuperator}} + M_{\mathrm{reactor}} \tag{26}$$

The total cycle system mass excludes piping, working fluids, and auxiliary components, considering only heat exchangers and turbo machinery assemblies. Both the cooler and heat exchanger employ PCHEs. Their mass models are calculated as follows:

$$M_{\mathrm{PCHE}} = \rho(V_{\mathrm{PCHE}} - A_c L) \tag{27}$$

Where the V_{PCHE} is Volume of PCHE. The total cross-sectional area of the recuperator, combined with its total length, can then be used to calculate the V_{PCHE}. A_c is the fluid channel total cross-sectional area of PCHEs.

Based on historical prototype development and concepts from the Space Station Freedom (SSF) project, NASA Glenn Research Center established a logarithmic curve-fitting relationship for compressor or turbine mass, expressed by the following formula [10]:

$$M_{\mathrm{trubin\&compressor}} = A\sqrt{0.5\cdot\gamma}\cdot[30.522\log(W_e) - 5.7178] \tag{28}$$

Where $W_e$is output or input power, the A and $\gamma$is fit coefficient. The A is 0.4–0.8, the γ is 1.3.

The mass of the reactor can be calculated using the following formula [21]:

$$M_{\mathrm{Reactor}} = 0.2195 Q_{\mathrm{Reator}} + 98.36 \tag{29}$$

Where $Q_{\mathrm{reactor}}$is output power of reactor.

3.5. Model verification

The validation of thermo-physical properties model of He-Xe mixture based on the semi-empirical correlations proposed by Tournier and El-Genk [37] and the verification of the thermodynamic model of the He-Xe Brayton cycle are presented here to ensure the reliability of the results for this study. The variation of the thermo-physical properties of He-Xe mixture with mole fraction under different temperatures is shown in Fig. 7. It can be seen that the thermo-physical properties predicted by this study agree well with the reference data, indicating that the thermo-physical properties model of He-Xe mixture developed in this work is reliable. In addition, the comparison in thermodynamic performance of

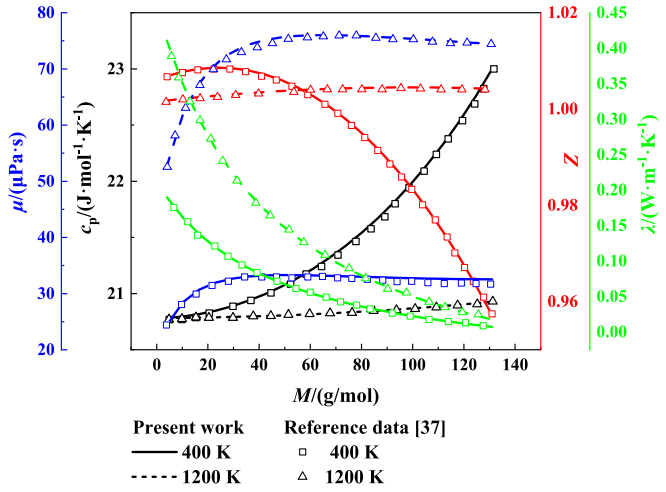


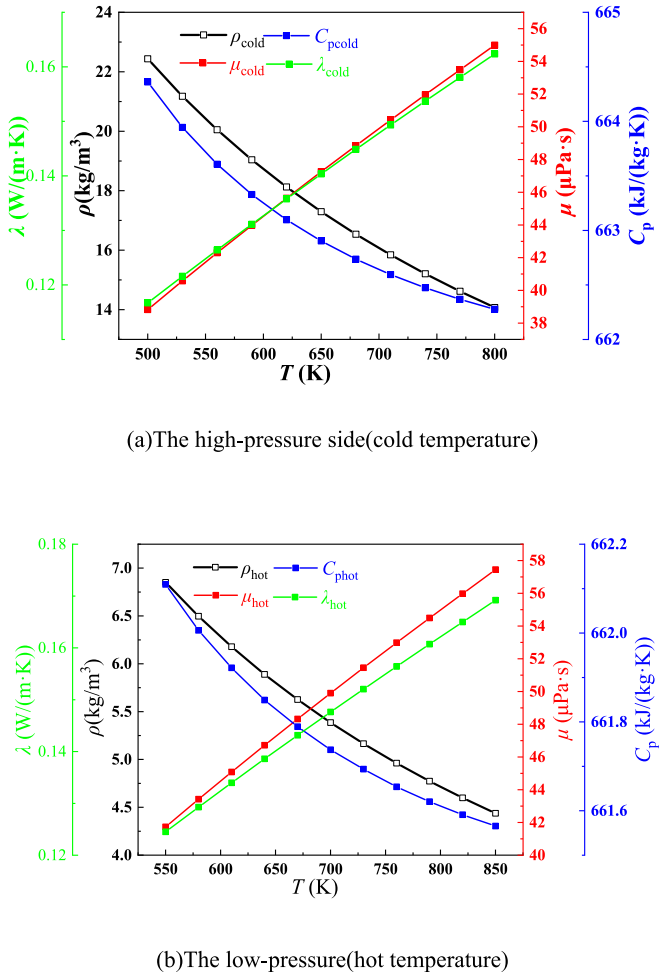
Fig. 7. The thermophysical properties and the verification of the He-Xe mixture.

Table 1
Model validation of He-Xe power cycle.

Parameters	Value		
Compressor outlet pressure(MPa)	2		
Compressor inlet pressure(MPa)	1.143		
Cycle power(KW)	30.1		
Coolant bleed line	2%		
Generator efficiency	0.9		
Generator Mechanical efficiency	0.867		
turbine efficiency	0.875		
Compressor efficiency	0.83		
Recuperator efficiency	0.95		
turbine inlet temperature(K)	1144		
Compressor inlet temperature(K)	403		
	Ref. [38]	This work	Error
turbine outlet temperature(K)	962	960	0.21%
Compressor outlet temperature(K)	525.5	528	0.47%
Power of the cooler(KW)	106.7	108	1.20%
Power of the heater(KW)	143	145.7	1.85%
mass flow rate(kg/s)	1.344	1.345	0.07%
power generation efficiency(%)	21.8	21.4	1.87%

Table 2
The basic input parameters of the He-Xe power cycle.

Parameters	Values
External diameter of fuel element(mm)	16.82
Pitch-to-diameter ratio of fuel element(P/D)	1.25
Maximum linear power density of fuel element(W/m)	59,000
Fuel length(mm)	15
Fuel arrangement layer count	11
Total core heat(kW)	351.345
Compressor pressure ratio	2.4/3.0
Compressor inlet pressure(MPa)	1
Coolant bleed line	5%
Generator efficiency	0.9
Generator Mechanical efficiency	0.867
Turbine efficiency	0.9
Compressor efficiency	0.85
Recuperator efficiency	0.95
turbine inlet temperature(K)	1150
Compressor inlet temperature(K)	400



(a)The high-pressure side(cold temperature)

(b)The low-pressure(hot temperature)

Fig. 8. The variations in He-Xe thermo physical properties with temperature under high-pressure and low-pressure conditions.

the recuperated He-Xe Brayton cycle between this work and reference data is displayed in Table 1. It can be found that the maximum error of thermodynamic parameters is lower than 2%, confirming that the thermodynamic model of the He-Xe Brayton cycle is reliable.

4. Results and discussion

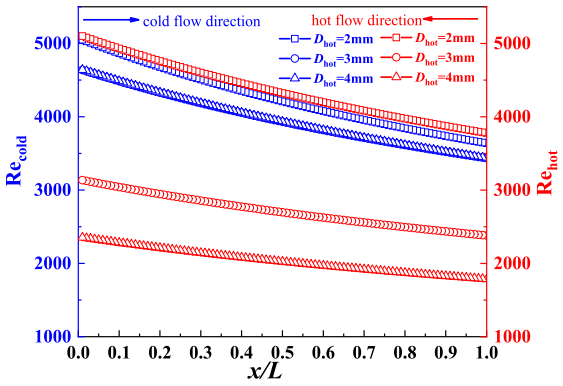
In this section, the effects of recuperator channel diameter and channel count on both recuperator performance and cycle performance are investigated under identical core structures and heat output conditions. Furthermore, cycle performance comparisons between Cycle 1 and Cycle 2 are presented. The basic input parameters of the He-Xe power cycle are given in Table 2. It should be noted that Table 1 only demonstrates the accuracy of the calculation method for the operating system, while the operating conditions in Table 2 are not related to those in Table 1.Fig. 8 presents schematic diagrams of the variations in He-Xe thermo physical properties with temperature under high-pressure and low-pressure conditions. Specifically, Fig. 8(a) shows the property variations under high-pressure conditions, and Fig. 8(b) shows the property variations under low-pressure conditions.

4.1. The effect of the hot fluid channel diameter

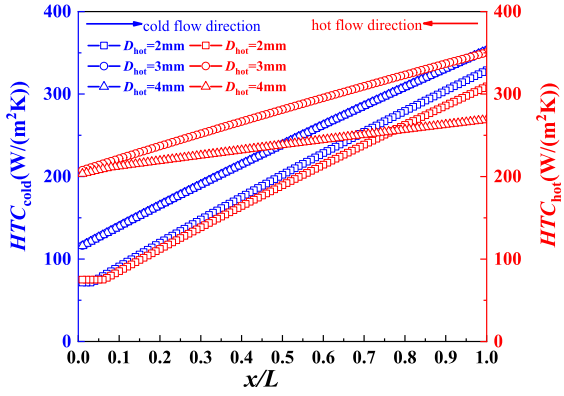
This subsection investigates the effect of the hot fluid channel diameter (D_{hot}) on the overall performance of the He-Xe power cycle and its recuperator. It should be noted that the cold fluid channel diameter (D_{cold}) differs from D_{hot} in the recuperator design. The basic input parameters for the He—Xe power cycle are provided in Table 2. The analysis considers D_{hot} values ranging from 2 mm to 4 mm, while maintaining a constant total channel number (N) of 6000 and a fixed cold D_{cold} of 2 mm.

Fig. 9(a) illustrates the effect of D_{hot} on the Reynolds number (Re). It can be observed that as D_{hot} decreases, Re_{hot} increases significantly, while Re_{cold} remains essentially unchanged. Furthermore, the figure shows that both Re_{hot} and Re_{cold} inside the recuperator decrease with rising temperature. This is due to the decrease in the density ($\rho_{He\text{-}Xe}$) of the helium-xenon working fluid with increasing temperature, coupled with an increase in its dynamic viscosity ($\mu_{He\text{-}Xe}$), as shown in Fig. 8. Generally, the variation in the heat transfer coefficient (HTC) is primarily influenced by changes in Re. Therefore, when D_{hot} exceeds 3 mm, the HTC increases as D_{hot} decreases, as depicted in Fig. 9(b). For a constant channel diameter, the HTC inside the recuperator increases with temperature, which is caused by the rise in the thermal conductivity (λ) of the helium-xenon working fluid with temperature, as shown in Fig. 8. It is important to note that at larger D_{hot} values, due to the lower Re-even reaching the laminar flow regime-the heat transfer performance is poor; consequently, the HTC decreases with increasing temperature. Ultimately, the overall heat transfer coefficient (U) increases with increasing D_{hot}.

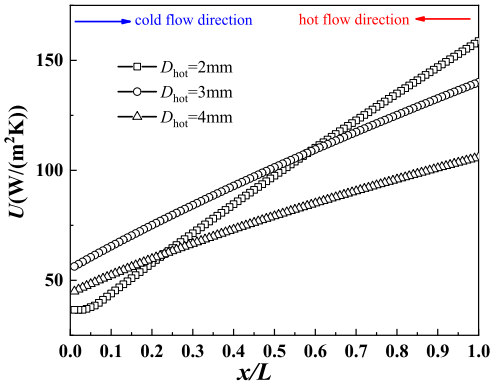
Increasing the channel diameter raises the height of the recuperator, as shown in Fig. 10(a). Under constant mass flow rate, a larger diameter reduces the Reynolds number (Re), thereby degrading heat transfer performance. This necessitates an increase in heat transfer area and recuperator length. As the length, width, and height increase



a. The effect of recuperator hot channel diameter on the variation of Re along flowing direction



b. The effect of recuperator hot channel diameter on the variation of *HTC* along flowing direction

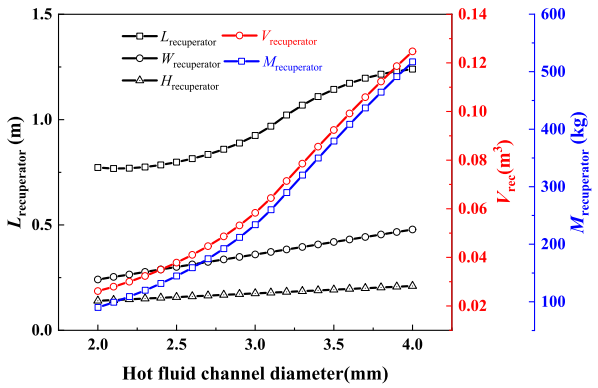


c. The effect of recuperator hot channel diameter on the variation of *U* along flowing direction

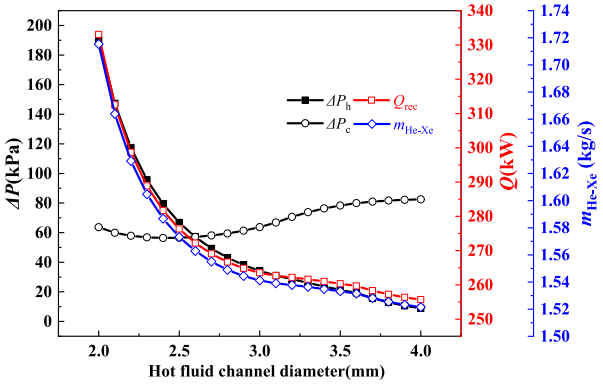
Fig. 9. The *Re*, *HTC*, *U* of the hot fluid channel diameter on the recuperator.

simultaneously, the recuperator volume ($V_{recuperator}$) and mass ($M_{recuperator}$) also increase accordingly.

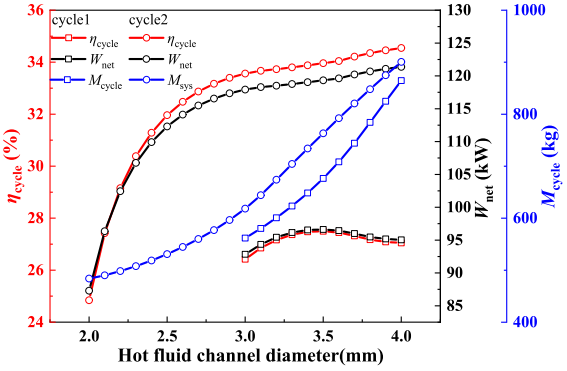
Fig. 10(b) shows the variations of recuperator pressure drop (ΔP), heat transfer rate ($Q_{recuperator}$), and He-Xe mass flow rate ($m_{He\text{-}Xe}$) with D_{hot}. The figure indicates that $Q_{recuperator}$ and $m_{He\text{-}Xe}$ are closely correlated, both decreasing as D_{hot} increases. The reduction in $m_{He\text{-}Xe}$ decreases the flow velocity ($v_{He\text{-}Xe}$) on both sides, leading to a lower ΔP, though this effect is partially offset by the increase in recuperator length. Furthermore, the increase in D_{hot} specifically reduces the flow velocity on the hot side, resulting in a decrease in hot-side pressure drop (ΔP_h), while the cold-side pressure drop (ΔP_c) remains unchanged. It can also be seen from the figure that ΔP_h is significantly higher than ΔP_c. This is mainly because the He-Xe on the hot side is a low-pressure gas with lower density, leading to a higher flow velocity under the same cross-



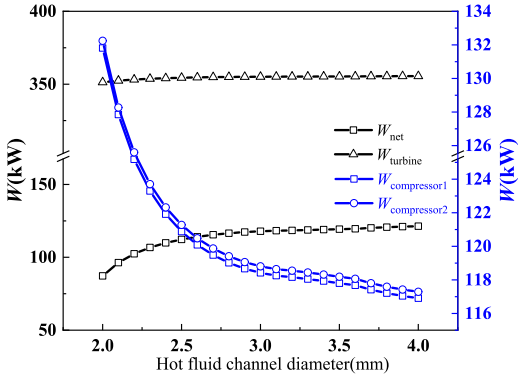
a. Variation of length, width, heght, bulk and mass of recuperator



b. Variation of ΔP_h, ΔP_c, $Q_{recuperator}$ and $m_{He\text{-}Xe}$



d. Thermodynamic performance and mass of the He-Xe power cycle



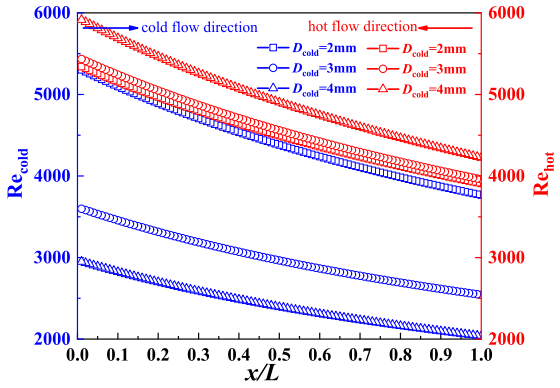
c. The variation of W_{net}, $W_{turbine}$, $W_{compressor}$

Fig. 10. The effect of the hot fluid channel diameter on the overall performance and recuperator of He-Xe power cycle.

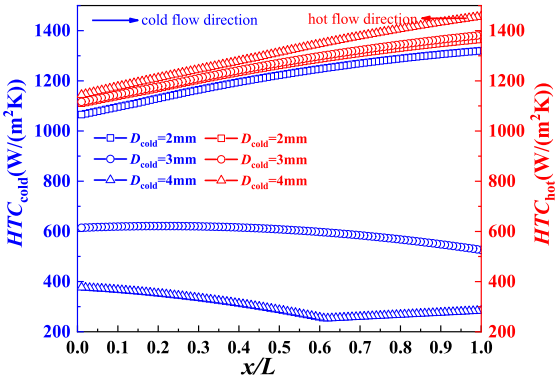
sectional area, which in turn increases the pressure loss on the hot side.

The decrease in $m_{He\text{-}Xe}$ reduces the compressor power consumption ($W_{compressor}$). Although a lower $m_{He\text{-}Xe}$ also reduces the turbine output, the decrease in ΔP increases the turbine expansion ratio, resulting in an almost unchanged net turbine output work ($W_{turbine}$). Consequently, the net cycle output work (W_{net}) increases with D_{hot}, as shown in Fig. 10(c).

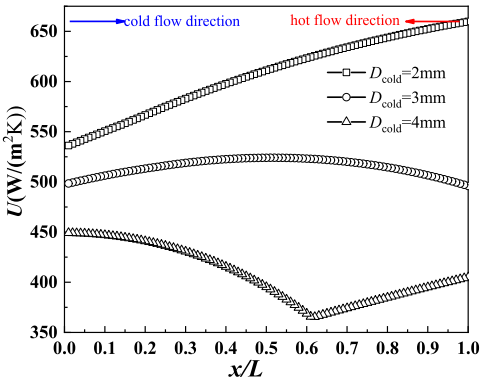
Fig. 10(d) compares the variations of thermal efficiency (η_{cycle}), net output work (W_{net}), and total system mass (M_{sys}) with D_{hot} for Cycle 1 and Cycle 2. As D_{hot} increases, $M_{recuperator}$ increases while the heater heat transfer rate (Q_{heater}) remains constant. Considering that for Cycle 1, a lower D_{hot} results in a case where the turbine output work is lower than the compressor input work, the comparison for Cycle 1 is only made for D_{hot} ranging from 3 mm to 4 mm. For Cycle 1: M_{sys} increases from 561.74 kg to 864.58 kg, W_{net} increases from 92.84 kW to 96.44 kW, and η_{cycle} increases from 26.42% to 27.50%. For Cycle 2: M_{sys} increases from 483.94 kg to 900.51 kg, W_{net} increases from 87.29 kW to 121.37 kW, and η_{cycle} increases from 24.85% to 34.55%. Compared to Cycle 1, Cycle 2 demonstrates superior thermodynamic performance, albeit with



a. The effect of recuperator cold channel diameter on the variation of Re along flowing direction



b. The effect of recuperator cold channel diameter on the variation of *HTC* along flowing direction



c. The effect of recuperator cold channel diameter on the variation of *U* along flowing direction

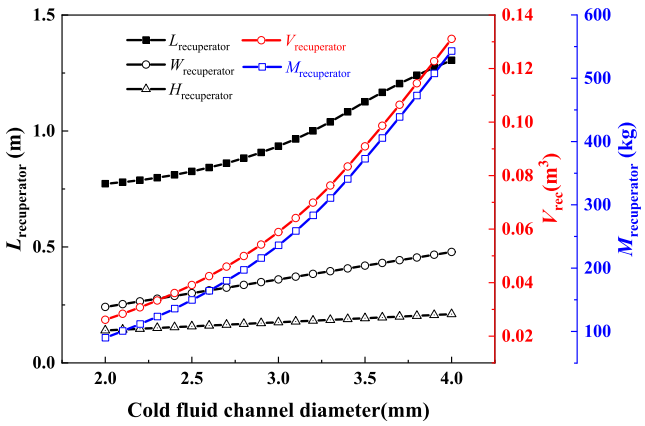
Fig. 11. The *Re*, *HTC*, *U* of the cold fluid channel diameter on the recuperator.

a slight increase in total cycle mass.

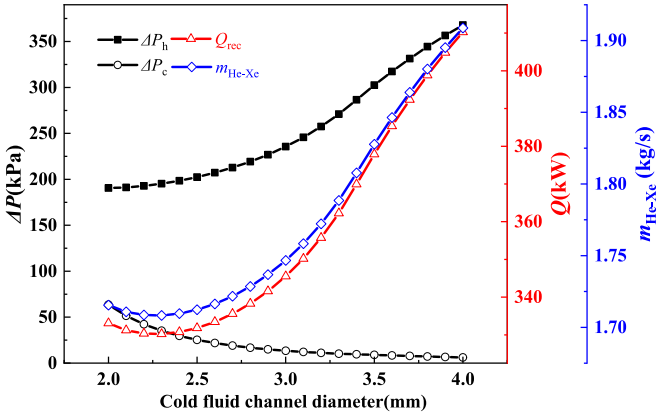
4.2. The effect of the cold fluid channel diameter

Here, we investigate the influence of D_{cold} on the recuperator structure and system performance. The study assumes a fixed D_{hot} of 2 mm and 6000 channels on both the cold and hot sides, while D_{cold} varies from 2 mm to 4 mm.

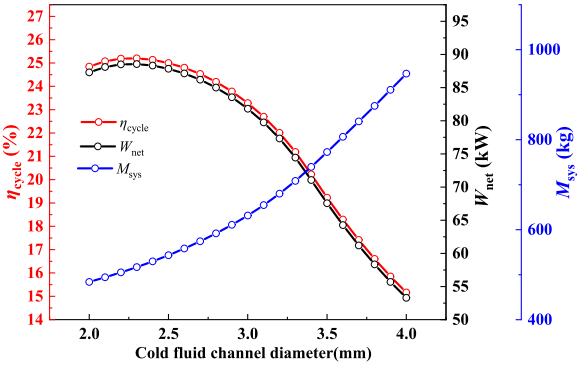
Fig. 11(a) shows the variation of Reynolds number (*Re*) with D_{cold}. It can be seen from the figure that as D_{cold} decreases, the cold-side Reynolds number (Re_{cold}) increases significantly, while the hot-side Reynolds number (Re_{hot}) changes little. Meanwhile, inside the recuperator, Re decreases with increasing temperature due to the changes in the properties of He-Xe with temperature. As shown in Fig. 11(b), as D_{cold}



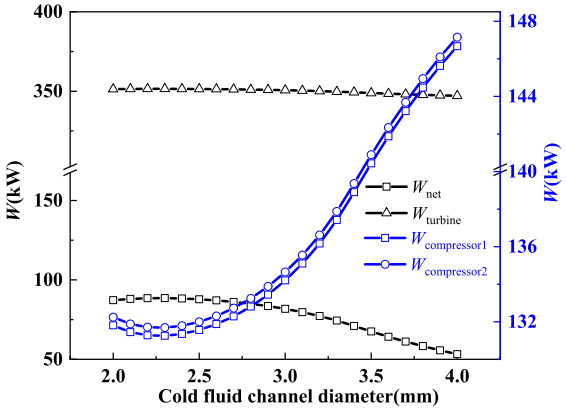
a. Variation of length, width, heght, bulk and mass of recuperator



b. Variation of ΔP_h, ΔP_c, $Q_{recuperator}$ and $m_{He\text{-}Xe}$



d. Thermodynamic performance and mass of the He-Xe power cycle



c. The variation of W_{net}, $W_{turbine}$, $W_{compressor}$

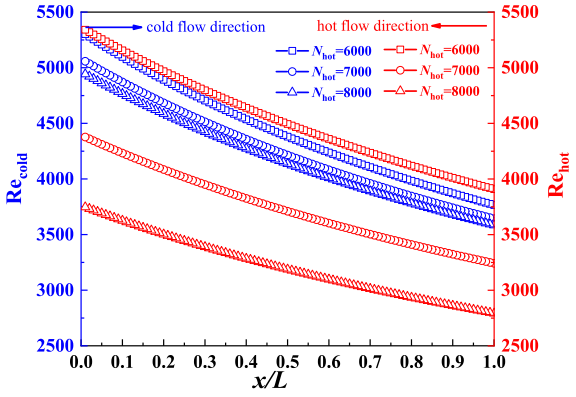
Fig. 12. The effect of the cold fluid channel diameter on the overall performance and recuperator of He-Xe power cycle.

decreases, the cold-side heat transfer coefficient (HTC_{cold}) correspondingly increases, which is caused by the increase in Re_{cold}. Meanwhile, with the channel diameter held constant, the *HTC* inside the recuperator increases with temperature, which is caused by the increase in the thermal conductivity (λ) of the helium-xenon working fluid with temperature, as shown in Fig. 8. It is important to note that at larger D_{cold} values, due to the lower *Re*-even reaching the laminar flow regime-the heat transfer performance is poor; therefore, as temperature increases,

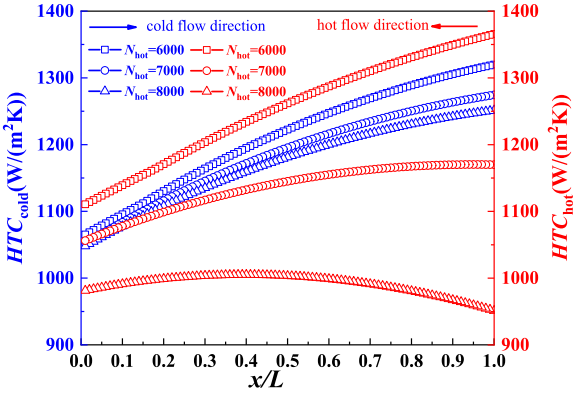
HTC instead decreases. Finally, the overall heat transfer coefficient (*U*) increases with increasing D_{cold}.

Similarly, increasing D_{cold} leads to an increase in the length, width, height, volume, and mass of the recuperator, as shown in Fig. 12(a). Fig. 12(b) illustrates the variations of ΔP, $Q_{recuperator}$, and $m_{He\text{-}Xe}$ with D_{cold}. It can be observed from the figure that as D_{cold} increases, both $m_{He\text{-}Xe}$ and $Q_{recuperator}$ first decrease and then increase. Due to the combined effect of the increasing recuperator length and the initial decrease followed by an increase in $m_{He\text{-}Xe}$, ΔP_h increases with increasing D_{cold}. Meanwhile, as the increase in D_{cold} enlarges the cold-side channel cross-sectional area and reduces $v_{He\text{—}Xe}$, ΔP_c decreases with increasing D_{cold}.

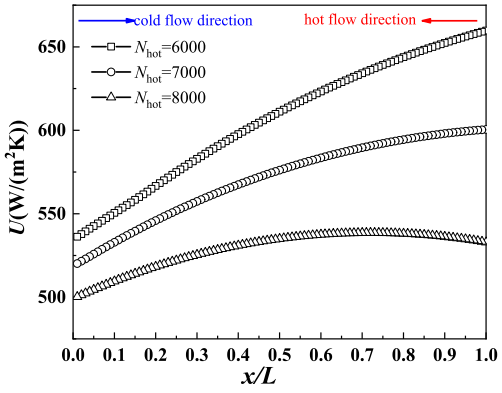
With the variation of $m_{He\text{-}Xe}$, both $W_{compressor1}$ and $W_{compressor2}$ show



a. The effect of recuperator hot channel number on the variation of Re along flowing direction



b. The effect of recuperator hot channel number on the variation of *HTC* along flowing direction



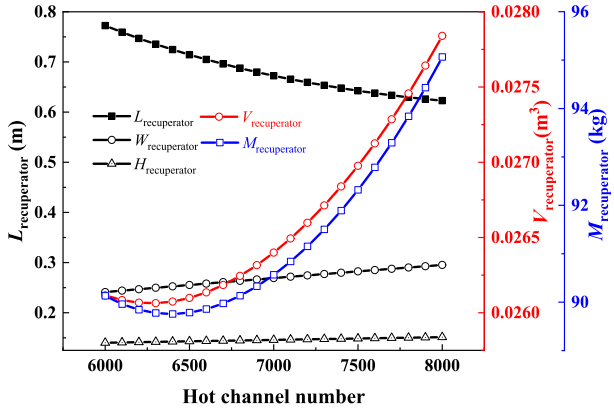
c. The effect of recuperator hot channel number on the variation of *U* along flowing direction

Fig. 13. The *Re*, *HTC*, *U* of the hot fluid channel number on the recuperator.

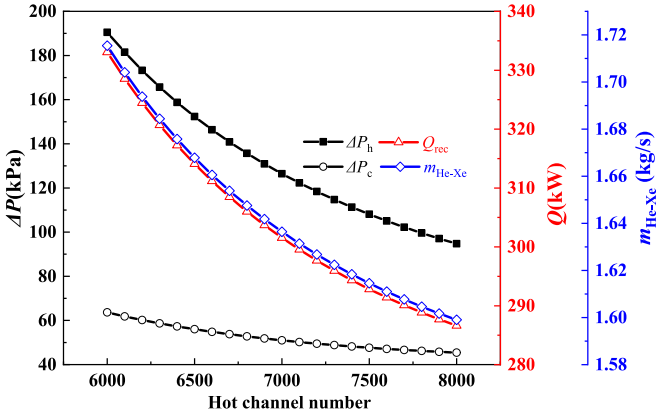
a trend of first decreasing and then increasing. Due to the increase in ΔP_h, the turbine expansion ratio decreases, resulting in Wturbine remaining unchanged, as shown in Fig. 12(c). Because of the lower number of hot-side channels and smaller channel diameter, Cycle 1 exhibits higher pressure losses; therefore, a comparison of cycle performance is not conducted in this section. Fig. 12(d) shows the variations of η_{cycle}, W_{net}, and M_{sys} for Cycle 2 with D_{cold}. It can be seen from the figure that η_{cycle} and Wnet first increase and then decrease as Dcold increases, while M_{sys} increases with increasing D_{cold}. The maximum values for η_{cycle} and W_{net} are 25.20% and 88.55 kW, respectively, while M_{sys} increases from 483.93 kg to 946.84 kg.

4.3. The effect of the hot channel number

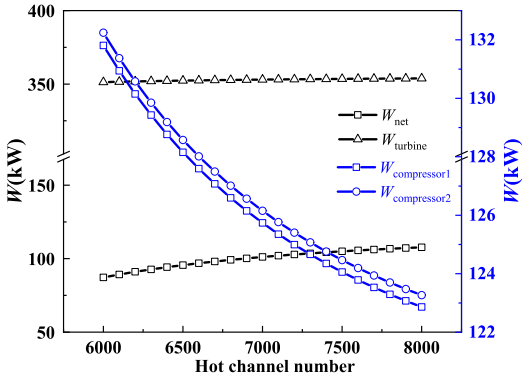
The following segment investigates the influence of the hot-side channel number (N_{hot}) on the recuperator structure and system



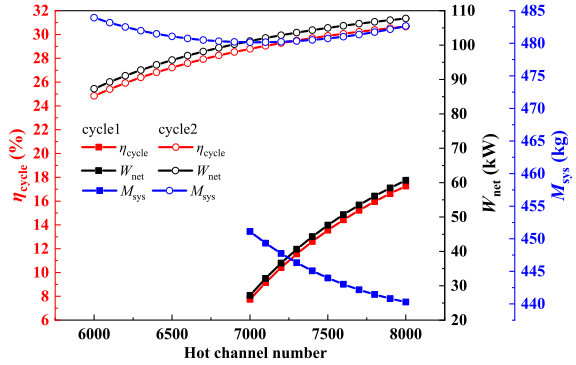
a. Variation of length, width, heght, bulk and mass of recuperator



b. Variation of ΔP_h, ΔP_c, Q_{rec} and m_{He-Xe}



c. The variation of W_{net}, $W_{turbine}$, $W_{compressor}$

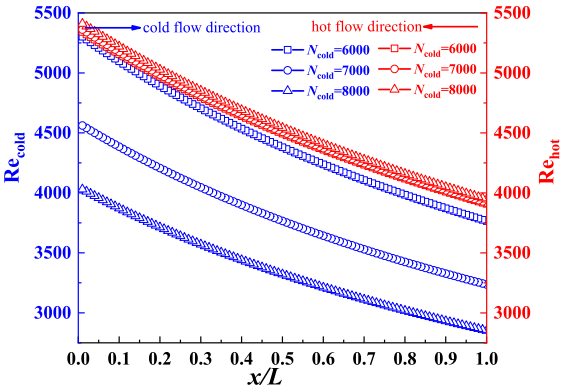


d. Thermodynamic performance and mass of the He-Xe power cycle

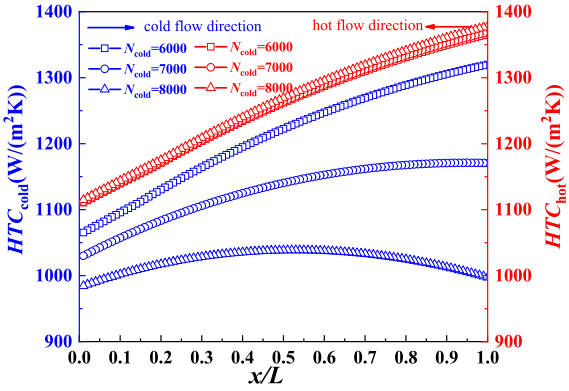
Fig. 14. The effect of the hot channel number on the overall performance and recuperator of He-Xe power cycle.

performance, with N_{cold} fixed at 6000, channel diameters maintained at 2 mm for both sides, and N_{hot} varying from 6000 to 8000.

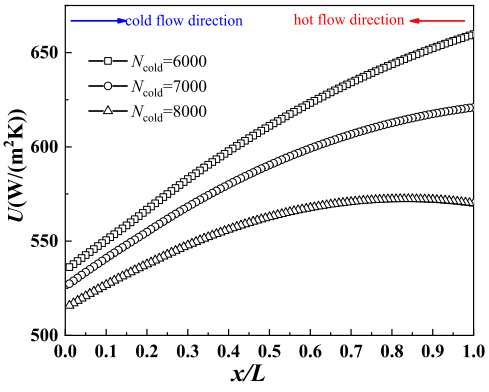
Fig. 13(a) presents the variation of *Re* on both sides of the recuperator with N_{hot}. Similar to changing the channel diameter, when N_{hot} decreases, Re_{hot} increases significantly due to the increased fluid velocity on the hot side, while Re_{cold} increases slightly. Meanwhile, the increase in Re_{hot} leads to an increase in HTC_{hot}, as shown in Fig. 13(b). Furthermore, inside the recuperator, rising temperature alters $\rho_{He\text{-}Xe}$ and $\mu_{He\text{-}Xe}$. Therefore, inside the recuperator, as temperature increases, the *HTC* on both sides increases despite the decrease in *Re*. It should be noted that at higher N_{hot}, where *Re* is lower (even reaching laminar flow), the heat transfer performance is poor, and *HTC* instead decreases with increasing temperature. The change in *HTC* further affects the overall heat transfer coefficient *U*; consequently, as N_{hot} decreases, the



a. The effect of recuperator cold channel number on the variation of Re along flowing direction



b. The effect of recuperator cold channel number on the variation of *HTC* along flowing direction

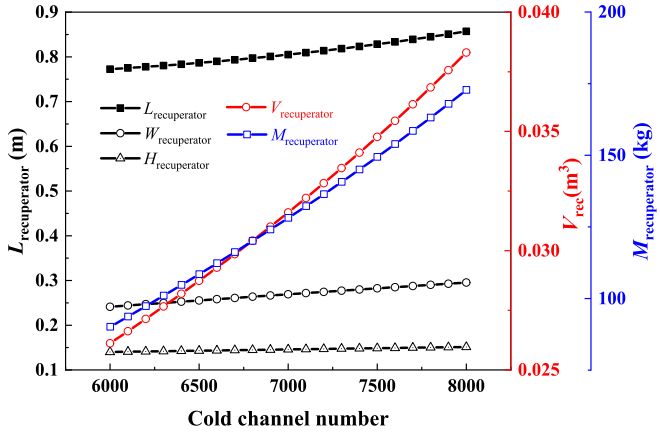


c. The effect of recuperator cold channel number on the variation of *U* along flowing direction

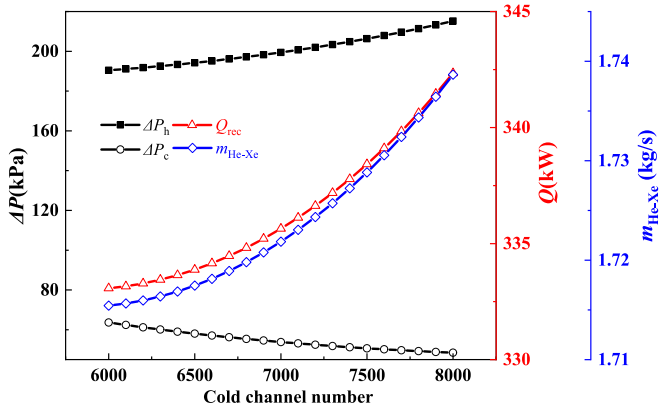
Fig. 15. The *Re*, *HTC*, *U* of the cold fluid channel number on the recuperator.

trend of U follows the same pattern as HTC_{hot}, as shown in Fig. 13(c).

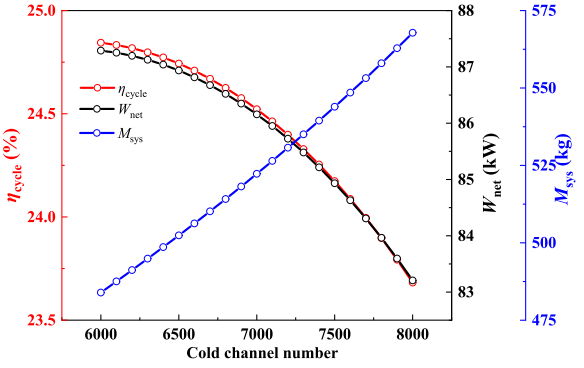
The channels of the recuperator studied in this paper have the same arrangement in both the longitudinal and transverse directions. Therefore, when the number of channels is increased, the total length and total width of the recuperator increase, as shown in Fig. 14(a). Obviously, increasing the number of channels under constant mass flow rate reduces the fluid velocity, so the hot-side $v_{recuperator}$ decreases as N_{hot} increases. This lowers the Re of the hot-side fluid, leading to reduced heat transfer effectiveness and an increased required heat transfer area. Since increasing N_{hot} significantly enlarges the wetted perimeter around the



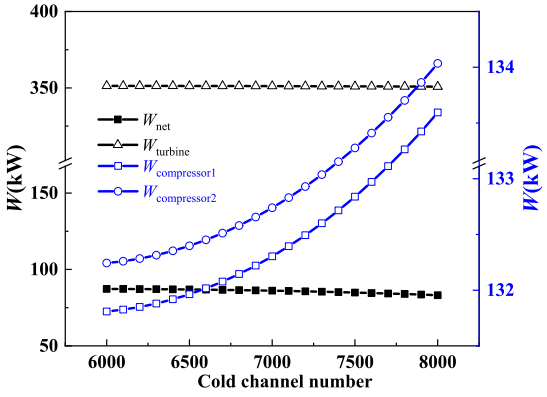
a. Variation of length, width, heght, bulk and mass of recuperator



b. Variation of ΔP_h, ΔP_c, Q_{rec} and $m_{He\text{-}Xe}$



d. Thermodynamic performance and mass of the He-Xe power cycle



c. The variation of W_{net}, $W_{turbine}$, $W_{compressor}$

Fig. 16. The effect of the cold channel number on the overall performance and recuperator of He-Xe power cycle.

channels, $L_{recuperator}$ decreases with increasing N_{hot}. Due to the reduction in $W_{recuperator}$ and $H_{recuperator}$, both $M_{recuperator}$ and $V_{recuperator}$ show a trend of first decreasing and then increasing with N_{hot}.

Fig. 14(b) presents the variations of ΔP_h, ΔP_c, $Q_{recuperator}$, and $m_{He\text{-}Xe}$ with N_{hot}. It can be seen from the figure that $m_{He\text{-}Xe}$ decreases as N_{hot} increases, and consequently $Q_{recuperator}$ also decreases. Considering the reduction in $L_{recuperator}$ and $m_{He\text{-}Xe}$, and since increasing the number of channels enlarges the total flow cross-sectional area, the pressure drops on both sides of the recuperator decrease with increasing N_{hot}.

Similarly, the reduction in $m_{He\text{-}Xe}$ and ΔP_h leads to a decrease in $W_{compressor}$, while $W_{turbine}$ remains unchanged, as shown in Fig. 14(c). Fig. 14(d) compares the variations of η_{cycle}, W_{net}, and M_{sys} with N_{hot} for Cycle 1 and Cycle 2. Considering that for Cycle 1, a lower N_{hot} results in a case where the turbine output work is lower than the compressor input work, the comparison for Cycle 1 is only made for N_{hot} ranging from 7000 to 8000. As N_{hot} increases: For Cycle 1: M_{sys} decreases from 451.10 kg to 440.30 kg, W_{net} increases from 27.23 kW to 60.67 kW, and η_{cycle} increases from 7.75% to 17.27%. For Cycle 2: M_{sys} decreases from 483.93 kg to 480.51 kg, W_{net} increases from 87.29 kW to 107.75 kW, and η_{cycle} increases from 24.84% to 30.67%. From these results, it can be concluded that increasing N_{hot} can significantly improve the thermodynamic performance of the cycle system, while the change in total mass is small.

4.4. The effect of the cold channel number

This part systematically examines the influence of N_{cold} on recuperator configuration and system performance. The analysis considers N_{cold} values ranging from 6000 to 8000, while maintaining a constant N_{hot} of 6000 and uniform channel diameters of 2 mm for both sides.

Fig. 15(a) shows the variation of Reynolds number (*Re*) inside the recuperator with N_{cold}. Similarly, when N_{cold} decreases, the cold-side fluid velocity increases, leading to an increase in Re_{cold}, while Re_{hot} increases slightly. The heat transfer coefficient (*HTC*) is closely related to *Re*; therefore, decreasing Ncold also increases HTC_{cold}. Inside the recuperator, as the temperature increases, the *HTC* on both sides increases accordingly. It is noteworthy that at higher N_{cold}, due to the lower *Re* (even reaching laminar flow), the heat transfer performance is poor, and *HTC* instead decreases with increasing temperature, as shown in Fig. 15(b). The change in *HTC* further affects the overall heat transfer coefficient *U*; therefore, the variation of the overall heat transfer coefficient (*U*) with N_{cold} follows the same pattern as the variation of *HTC* with temperature, as shown in Fig. 15(c).

Fig. 16(a) shows that increasing N_{cold} raises both the transverse and longitudinal arrangement counts and reduces the cold-side heat transfer effectiveness, leading to corresponding increases in $L_{recuperator}$, $W_{recuperator}$, and $H_{recuperator}$. Clearly, the increases in $L_{recuperator}$, $W_{recuperator}$, and $H_{recuperator}$ result in larger $V_{recuperator}$ and $M_{recuperator}$.

From Fig. 16(b), it can be observed that $Q_{recuperator}$ and $m_{He\text{-}Xe}$ increase with increasing N_{cold}. Due to the combined effect of longer recuperator length and higher $m_{He\text{-}Xe}$, ΔP_h rises with increasing N_{cold}. Meanwhile, because increasing N_{cold} expands the cold-side channel cross-sectional area and lowers $v_{He—Xe}$, ΔP_c decreases as N_{cold} increases.

With the pressure ratio held constant, the increase in $m_{He\text{-}Xe}$ causes both $W_{compressor1}$ and $W_{compressor2}$ to rise. However, due to the increase in ΔP_h, the turbine expansion ratio decreases, so $W_{turbine}$ remains unchanged, as shown in Fig. 16(c).

Because of the lower number of hot-side channels and smaller channel diameter, Cycle 1 exhibits higher pressure losses; therefore, no cycle performance comparison is conducted in this section. Fig. 16(d) presents the variations of η_{cycle}, W_{net}, and M_{sys} for Cycle 2 with N_{cold}. It can be seen that as $W_{compressor1}$ and $W_{compressor2}$ increase while $W_{turbine}$ stays constant, W_{net} decreases. Consequently, the results show that with increasing N_{cold}: η_{cycle} for Cycle 2 decreases from 24.84% to 23.68%, W_{net} decreases from 87.29 kW to 83.26 kW, while M_{sys}increases from 483.93 kg to 567.83 kg. It can be concluded that increasing N_{cold} raises

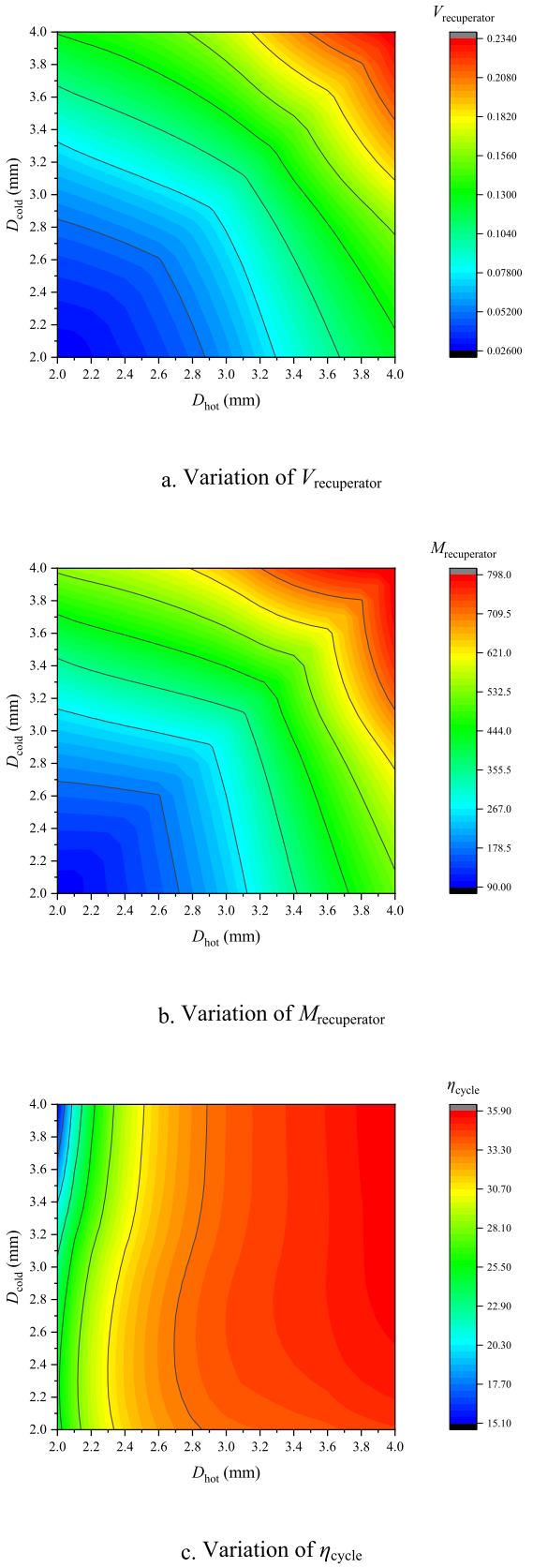


Fig. 17. Contour map of recuperator with various recuperator channel diameter.

the total mass of the cycle system and also reduces its thermodynamic performance.

4.5. The contour map of channel diameter versus channel number

This analysis employs contour map to visualize the variations of η_{cycle}, $M_{recuperator}$, and $V_{recuperator}$ with respect to channel diameter and channel number are presented. Fig. 17(a) shows relationship of $V_{recuperator}$ versus both hot-side and cold-side channel diameters, indicating that smaller values of D_{hot} and D_{cold} yield lower $V_{recuperator}$. Furthermore, reducing D_{hot} and D_{cold} also decreases $M_{recuperator}$, as depicted in Fig. 17 (b). It can also be observed from Fig. 17(a–b) that $V_{recuperator}$ and $M_{recuperator}$ are minimized when D_{hot} and D_{cold} are set equal. Although smaller channel diameters help reduce both $V_{recuperator}$ and $M_{recuperator}$, this comes at the expense of a decrease in η_{cycle}, as shown in Fig. 17(c). Specifically, as D_{hot} and D_{cold} increase, $V_{recuperator}$ rises from 0.0261m^3 to 0.2335m^3, $M_{recuperator}$ rises from 90.13 kg to 796.88 kg, η_{cycle} rises from 24.84%to 35.85%.

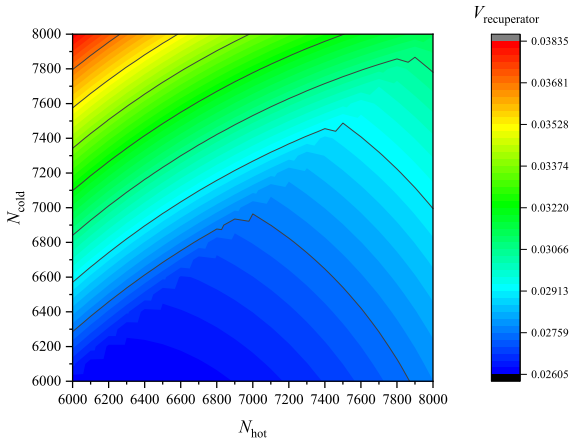
Fig. 18 presents contour map of $V_{recuperator}$, $M_{recuperator}$, and η_{cycle} under different combinations of N_{hot} and N_{cold}. As shown in Figs. 18(a) and 18(b), lower channel counts result in reduced values of both $V_{recuperator}$ and $M_{recuperator}$. The plots further reveal that when N_{cold} exceeds N_{hot}, $V_{recuperator}$ and $M_{recuperator}$ increase markedly with rising N_{cold}. Additionally, Fig. 18(c) indicates that increasing N_{hot} has a pronounced positive effect on η_{cycle}, which rises as N_{hot} increases. Specifically, the minimum and maximum values of both $V_{recuperator}$ and $M_{recuperator}$ occur at N_{hot} is 6000, with $V_{recuperator}$ increasing from 0.0261m^3 to 0.0383 m^3 and $M_{recuperator}$ from 90.13 kg to 172.80 kg as N_{cold} increases. Moreover, η_{cycle} improves from 24.84% to 30.67% with increasing N_{hot} and N_{cold}.

Based on the preceding analysis, it can be concluded that when the channel diameters and channel quantities on both sides of the recuperator are equal, the system exhibits superior performance in terms of $V_{recuperator}$, $M_{recuperator}$ and η_{cycle}. Accordingly, Fig. 19 presents contour map of $V_{recuperator}$, $M_{recuperator}$, and η_{cycle} under varying channel diameters and channel quantities. It should be noted that in Fig. 19, the channel diameters and channel quantities are set to be identical on both the cold and hot sides of the recuperator. As shown in Figs. 19(a) and 19 (b), smaller channel diameters and lower channel quantities lead to reduced $V_{recuperator}$ and $M_{recuperator}$. However, increasing channel diameter and channel number helps improve η_{cycle}, as illustrated in Fig. 19(c). Specifically, as both channel diameter and channel number increase, $V_{recuperator}$ rises from 0.0261m^3 to 0.2317 m^3, $M_{recuperator}$ increases from 90.13 kg to 785.78 kg, and η_{cycle} improves from 24.84% to 36.20%.

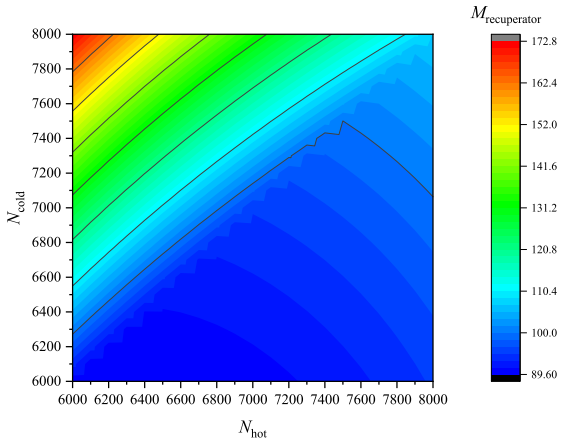
4.6. Multi-objective optimization

The He-Xe Brayton cycle applied to space stations requires minimal mass and superior thermodynamic performance. However, results indicate that recuperator design cannot simultaneously satisfy both conditions. Therefore, multi-objective optimization is necessary to derive the optimal design solution. In this study, the thermal power input from the heat source remains unchanged, making η_{cycle} the key metric for thermodynamic performance. By performing multi-objective optimization on M_{sys} and η_{cycle}, D_{hot}, N_{hot}, D_{cold}, and N_{cold} are selected as decision variables, with their variation ranges listed in Table 3. The Non-dominated Sorting Genetic Algorithm-II (NSGA-II) is adopted as the optimization method, and the corresponding program is implemented in MATLAB. For the NSGA-II method, the population size is set to 100, and the maximum number of generations is specified as 100.

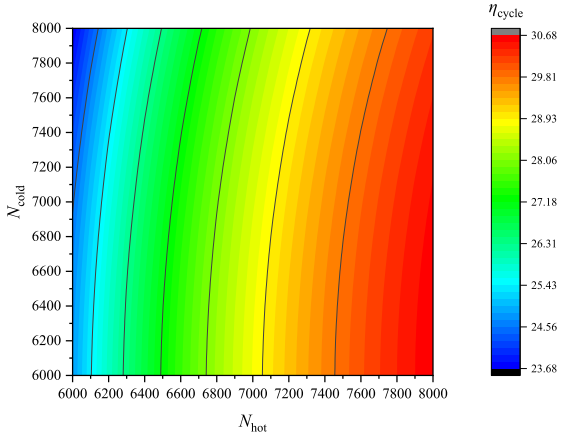
Fig. 20 presents the optimal Pareto frontier set obtained from bi-objective optimization. Each point in the figure can be regarded as an optimal design scheme. Decision-makers can select the final optimal solution by imposing constraints on these optimal points according to different practical requirements. The figure indicates that a lower M_{sys} requires acceptance of a lower η_{cycle}. Therefore, it is impossible to design



a. Variation of $V_{recuperator}$



b. Variation of $M_{recuperator}$



c. Variation of η_{cycle}

Fig. 18. Contour map diagrams of recuperator with various recuperator channel number.

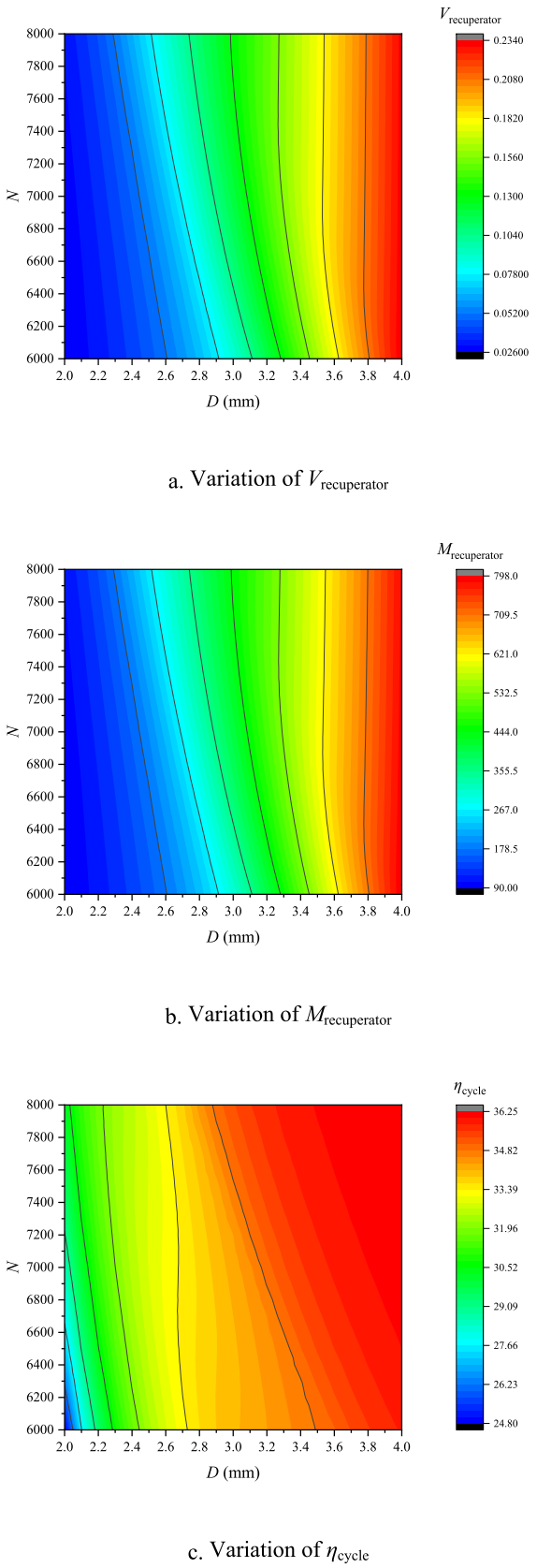


Fig. 19. Contour map diagrams of recuperator with various recuperator channel diameter and channel number.

Table 3
The variation range of the decision variables.

Parameters	Values
D_{hot}	2 mm–4 mm
N_{hot}	6000–8000
D_{cold}	2 mm–4 mm
N_{cold}	6000–8000

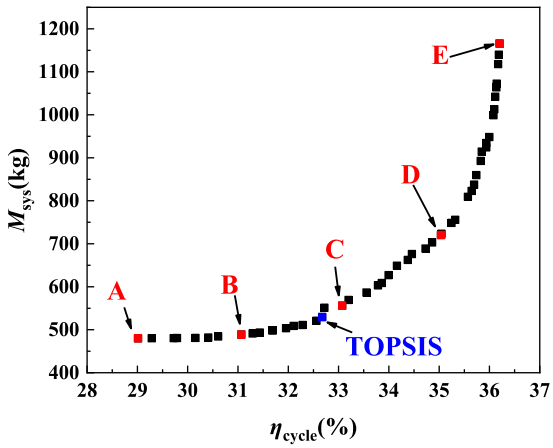


Fig. 20. Optimal Pareto front sets of two-objective optimizations.

a system that simultaneously achieves the minimum M_{sys} and the maximum η_{cycle}. Consequently, we employed the TOPSIS criterion [39,40]. Five schemes, labeled A through E, were randomly selected. Scheme A represents the smallest η_{cycle} and M_{sys}, while Scheme E represents the largest η_{cycle} and M_{sys}. The results of TOPSIS decision-making are also provided in the Table 4.

5. Conclusions

In this work, a comprehensive investigation was conducted on the design of the recuperator, including its internal structure, the geometric parameters and mass of the recuperator, the total mass of the He—Xe Brayton system on the space station, and the thermodynamic performance. Additionally, multi-objective optimization was carried out to obtain the optimal design of the recuperator. The main conclusions are as follows:

- Compared with the simple regenerative cycle, the intercooling cycle demonstrates superior performance in thermodynamic efficiency, and in terms of mass, it is only slightly higher than the simple regenerative cycle.
- Under the conditions of minimum channel diameter and minimum channel number, the recuperator can achieve minimum volume and mass, but it also leads to a reduction in cycle efficiency due to higher pressure losses. Therefore, during recuperator design, pressure loss should be considered, and the channel diameter and number should be reduced as much as possible while meeting low pressure loss requirements.
- Since the hot-side He-Xe is the lower-pressure gas in the cycle, under higher temperature and lower pressure conditions, the working fluid density is low, resulting in higher flow velocity under the same channel diameter and number, which leads to correspondingly higher pressure losses. Additionally, the high pressure loss on the hot side under low-pressure conditions has a greater impact on the turbine expansion ratio, thereby further reducing cycle efficiency. Therefore, in recuperator design, the cold-side channel diameter and number should be lower than those on the hot side.

Table 4
The optimal structural geometrical parameters of recuperator.

Parameters	Optimal desifgn points					
	A	B	C	D	E	TOPSIS
D_{hot}	3.5 mm	2.12 mm	2.79 mm	2.77 mm	4 mm	3.51
N_{hot}	7633	7931	6376	6374	8000	7634
D_{cold}	2.95 mm	2.01 mm	2.47 mm	2.46 mm	4 mm	3.05 mm
N_{cold}	6005	6129	6153	6154	6991	6000
M_{cycle}	480.22 kg	488.97 kg	556.15 kg	720.48gk	1165.74 kg	529.74 kg
η_{cycle}	29.01%	31.07%	33.07%	35.04%	36.20%	32.67%

- This study conducted multi-objective optimization of a He-Xe Brayton cycle for space stations using NSGA-II algorithm with recuperator parameters (D_{hot}, N_{hot}, D_{cold}, N_{cold}) as decision variables. Multiple optimal solutions are presented through the multi-objective optimization curve, while the best solution is selected using the TOPSIS decision-making method.

6. Design limitations and future work

The high-density design proposed in this study (channel diameter 2–4 mm, number up to 8000), while demonstrating excellent thermal-hydraulic performance, also presents several engineering challenges. First, in terms of manufacturing feasibility, such a large-scale microchannel array imposes extremely high demands on the uniformity and consistency of the diffusion bonding process, representing a primary technical bottleneck for industrialization. Second, regarding operational reliability, the risk of thermal stress concentration and fatigue failure in the thin-walled structures between channels under high-temperature-difference conditions requires further quantification through thermo-fluid-structure coupling analysis. Finally, with respect to long-term maintenance, microchannels are highly susceptible to fouling, necessitating the incorporation of online monitoring and chemical cleaning interfaces in the system design. These challenges constitute key research issues for advancing from experimental prototypes to engineering applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Fig. A1 presents the internal temperature distribution and the variation of temperature difference between the two sides within the recuperator

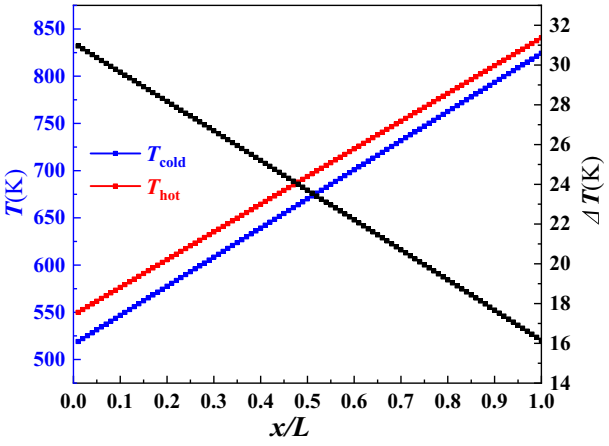
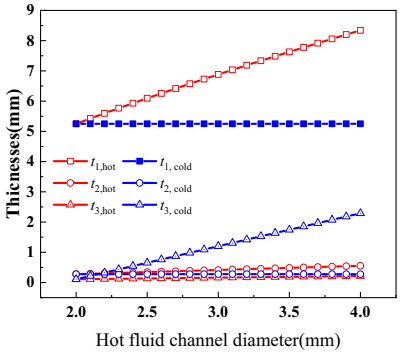
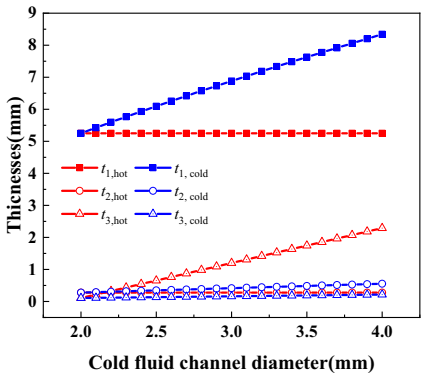


Fig. A1. Internal Temperature Distribution and Side-to-Side Temperature Difference in the recuperator.

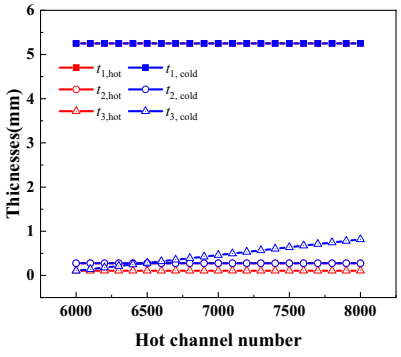
Fig. A2 illustrates variations in the internal structure of the heat exchanger with increasing D_{hot}, N_{hot}, D_{cold} and N_{cold}. The t_1 is the edge width, the t_2 is the channel diameter and the t_3 is the channel pitch.



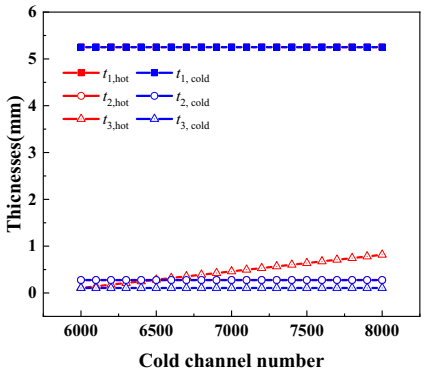
(a)Variation with D_{hot}



(b)Variation with N_{hot}



(c)Variation with D_{cold}



(a)Variation with N_{cold}

Fig. A2. The internal structure of the recuperator with increasing D_{hot}, N_{hot}, D_{cold} and N_{cold}.

Data availability

The data that has been used is confidential.

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